

Ameya Agarwal

Software Developer | Full-stack, Backend & AI/ML

+91-7078068986 | agarwal.ameya.5006@gmail.com | linkedin.com/in/ameya-agarwal | github.com/Ameya5006

Summary

Software Developer (B.Tech CSE, 2028) with hands-on experience in backend, full-stack and ML engineering. Owned backend for a Smart India Hackathon team and delivered a calibrated, leakage-safe healthcare ML pipeline; strong in Python, FastAPI, PostgreSQL, React and LLM function calling.

Education

Bennett University

B.Tech, Computer Science and Engineering — CGPA: 8.25 / 10

Greater Noida, Uttar Pradesh, India

2024 – 2028

Technical Skills

Languages: C++, Python, Java, JavaScript, TypeScript, Dart, HTML, CSS

AI/ML & LLM: scikit-learn, pandas, NumPy, Gemini API, Function Calling, Prompt Engineering, AI Guardrails

Backend: FastAPI, Node.js, Express.js, REST APIs, WebSockets, SQLAlchemy, Alembic, JWT, RBAC

Frontend & Mobile: React, Flutter, Vite, Tailwind CSS, Zustand, React Router

Data & Infrastructure: PostgreSQL, Supabase, MongoDB, Firebase Auth, Firestore, Docker, Nginx, Vercel, pytest, Apps Script

Professional Experience

Zarva (Early-stage Startup)

Remote

Full Stack Developer

Oct 2025 – Mar 2026

- Prototyped a **Flutter** mobile application with the founders, taking the product from requirements to a working demo.
- Developed features across Flutter interfaces and backend integrations, iterating rapidly as product requirements evolved.

Projects

SAHAY-AI — Real-time Trauma Assessment | *Python, FastAPI, WebSockets, PostgreSQL* | GitHub

- Led backend development for a **6-member** Smart India Hackathon team, building asynchronous REST/WebSocket workflows with JWT/RBAC, consent controls, ASR/TTS integration, role-filtered escalation, deterministic dialogue-safety gates and human-counsellor takeover for the National Helpline Against Atrocities.
- Migrated a **15-table** persistence layer from SQLite to Supabase PostgreSQL using SQLAlchemy, Alembic and TLS; validated the backend with **172 passing tests**, **35/35** scenario checks and **581 ms p95** server-side voice-turn latency.

Aletheia — Healthcare Risk Modelling & Audit System | *Python, scikit-learn, pandas, NumPy, pytest* | GitHub

- Built a reproducible healthcare ML pipeline across **99,343** hospital encounters and **69,990** patients, using patient-disjoint splits and grouped nested cross-validation to prevent identity leakage.
- Compared four model families and selected a sigmoid-calibrated Histogram Gradient Boosting model before a guarded holdout evaluation, achieving **0.203** average precision, **0.639** ROC-AUC and **0.098** Brier score, supported by **87 passing offline tests**.

PalmChef — Hands-free AI Cooking Assistant | *React, Node.js, MongoDB, Gemini API* | GitHub

- Engineered a bounded **Gemini function-calling** workflow exposing **13+** typed cooking tools for timers, recipe navigation, scaling, substitutions and generation through a maximum **3-round** execution loop.
- Added server-side function validation, browser-side allow-listed tool execution, recipe validation/repair and fallback model routing; protected the AI service with a **20-request/minute** limit, 5-minute session expiry and JWT-protected APIs.

GymFlow — Multi-gym Membership Platform | *React, TypeScript, Firebase, Apps Script* | GitHub | Live

- Launched a multi-tenant membership and payment platform on Vercel serving **2 gyms** and **30+ members**, with Firebase Authentication, Firestore and separate member/admin dashboards.
- Implemented gym-specific UPI payment flows and an asynchronous Apps Script automation layer that mirrors operational data to Google Sheets and sends daily WhatsApp membership-expiry reminders.

Leadership

Placement Committee Member | *Career Services Center, Bennett University*

Aug 2026 – Present

- Coordinate campus placement drives with **5+** recruiters, managing scheduling and communication affecting **500+** students.

Sub Head — Operations | *Bennett University Student Council*

Jul 2025 – May 2026

- Headed a **4-member** operations team coordinating 24-hour logistics for the 3-day Uphoria fest with **5,000+** attendees.

Core Team Member | *GeeksforGeeks Society*

- Organised **3** technical events and workshops, managing planning, documentation and execution for **150+** participants.

Achievements & Certifications

- **Smart India Hackathon 2026:** Secured **6th place among 591 teams**. **Sports:** District Boxing Silver, Fencing Bronze.
- **Certifications:** NPTEL Elite — Software Engineering, IIT Kharagpur (2025); Deloitte Australia Data Analytics Simulation.